

CHICAGO O'HARE, IL, USA

WMO: 725300

Lat: 41.960N

Lon: 87.932W

Elev: 202

StdP: 98.92

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94846

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.7	-16.0	-24.1	0.4	-18.0	-21.6	0.6	-15.2	11.9	-0.8	11.0	-1.4	4.7	270	0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	32.9	23.4	31.4	22.7	30.0	21.9	25.2	30.7	24.2	29.2	23.3	28.0	5.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	18.8	28.6	22.6	17.7	27.4	21.7	16.7	26.3	78.0	30.5	73.8	29.5	70.1	28.2	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	10.9	9.3	8.4	DB	-22.3	35.2	3.8	2.0	-25.1	36.7	-27.3	37.9	-29.5	39.0	-32.3	40.5
(5)				WB	-22.8	26.6	3.7	1.0	-25.4	27.3	-27.5	27.9	-29.6	28.4	-32.3	29.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	-4.4	-2.3	3.4	9.5	15.4	21.0	23.7	22.8	18.9	12.0	4.8	-1.3
	DBStd	10.99	6.54	6.31	6.10	5.06	4.94	4.12	3.25	2.93	4.18	4.89	5.35	5.95
	HDD10.0	1645	446	347	221	68	7	0	0	0	1	34	169	352
	HDD18.3	3421	704	579	464	268	121	21	2	2	43	203	406	608
	CDD10.0	1778	0	2	17	53	174	330	424	397	267	98	14	2
	CDD18.3	511	0	0	1	4	29	101	167	141	59	8	0	0
	CDH23.3	4498	0	0	11	48	293	957	1562	1082	484	63	0	0
	CDH26.7	1483	0	0	1	8	84	334	574	327	143	11	0	0
Wind	WSAvg	4.3	4.8	4.8	4.8	5.0	4.4	3.8	3.6	3.4	3.6	4.3	4.6	4.5
Precipitation	PrecAvg	908	51	47	59	85	103	104	97	101	73	76	63	49
	PrecMax	1175	117	154	128	185	190	266	205	228	262	213	152	117
	PrecMin	572	13	12	20	34	11	26	43	29	12	24	8	12
	PrecStd	136	28	29	28	36	43	51	40	49	51	49	38	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	16.9	24.8	27.7	31.6	33.9	35.8	33.7	32.6	28.3	20.6	16.0
		MCWB	12.0	12.3	17.2	17.3	20.9	23.2	24.9	24.3	22.6	20.5	14.2	13.9
	2%	DB	9.7	12.4	19.8	24.3	29.1	32.2	33.1	31.7	30.3	25.0	17.6	12.2
		MCWB	7.1	9.1	13.7	16.0	19.9	22.5	24.3	23.3	21.6	17.9	13.4	10.0
	5%	DB	6.2	8.6	16.0	21.6	26.7	30.3	31.4	30.0	28.1	22.5	15.6	9.1
		MCWB	4.2	5.6	11.5	14.1	18.6	21.8	23.1	22.4	20.4	16.7	12.0	7.0
	10%	DB	4.0	5.7	12.7	18.6	24.3	28.5	29.9	28.4	26.2	20.1	13.0	6.5
		MCWB	2.5	3.6	8.7	12.1	17.6	21.0	22.2	21.4	19.4	15.4	9.7	4.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.4	13.4	17.7	19.1	23.1	25.4	26.8	26.0	24.1	21.7	16.5	14.5
		MCDB	13.5	15.9	24.0	25.3	29.3	30.9	32.5	31.9	29.4	26.8	18.9	15.7
	2%	WB	7.8	9.3	14.9	16.7	21.3	24.0	25.6	24.6	22.9	19.6	14.5	10.5
		MCDB	9.1	12.5	18.6	22.4	26.6	29.8	31.0	29.5	27.9	22.7	16.9	11.7
	5%	WB	4.6	5.9	12.1	15.0	19.9	22.9	24.6	23.7	21.8	17.8	12.1	7.1
		MCDB	5.9	8.3	15.5	20.7	25.0	28.5	29.9	28.0	26.2	21.2	14.9	8.8
	10%	WB	2.6	3.7	8.7	12.8	18.4	21.9	23.6	22.8	20.6	15.9	10.1	4.6
		MCDB	4.0	5.6	12.4	17.9	23.0	27.1	28.4	26.7	24.8	19.5	12.8	6.4
Mean Daily Temperature Range	5% DB	MDBR	7.3	7.8	8.8	10.2	10.6	10.5	9.8	9.2	10.0	9.3	7.9	6.9
		MCDBR	9.3	11.5	13.2	15.2	13.5	12.1	11.0	10.8	12.1	12.7	11.1	9.2
		MCWBR	8.1	8.8	8.9	8.9	6.8	5.6	4.8	4.6	5.2	7.2	8.1	7.7
	5% WB	MCDBR	8.9	10.7	12.3	13.6	11.9	10.9	10.2	9.3	10.3	10.6	10.0	8.6
		MCWBR	8.2	8.7	9.2	9.1	7.1	6.0	5.3	4.8	5.1	7.4	8.3	7.9
Clear-Sky Solar Irradiance	taub	0.298	0.314	0.347	0.401	0.428	0.438	0.446	0.440	0.411	0.358	0.316	0.298	
	taud	2.435	2.383	2.382	2.252	2.230	2.249	2.228	2.242	2.291	2.439	2.507	2.470	
	Ebn at Noon	852	891	899	868	850	840	829	821	819	830	826	821	
	Edh at Noon	77	95	107	132	138	136	137	132	117	90	72	69	
All-Sky Solar Radiation	RadAvg	1.58	2.46	3.58	4.62	5.28	6.05	6.09	5.31	4.39	2.88	1.82	1.32	
	RadStd	0.18	0.31	0.28	0.38	0.43	0.41	0.30	0.31	0.37	0.32	0.18	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	-0.51	-0.60	N/A	N/A	N/A	N/A
(47)	Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page